

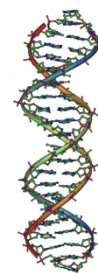
Outline of Lecture 2:

1. Encountered Data Types
2. Conceptual Introduction of Computer Programming
3. Introduction of Python
4. Basic Python Programming Mechanics

Encountered Data Types:

Different ways of analyzing data need different representations, depending on the structure of data.

- Sequential Data: A list of symbols or numbers that are in a certain order such as nucleic acid sequences.



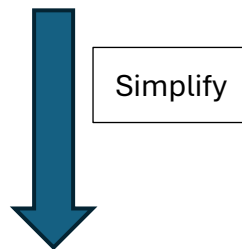
```
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CAGCGCGCTCTGAGGGAACAGTGAAGTCTACTTTGGGAATGGGTGAGCTACCG  
TGGCAGCAGACAGCTCAGCGAGTCGGTGGCTCTGGCTCCCGTGGAAATCCATGAT  
CTGATATGGCAGGTTTACACAGCAGACAGACCCGAGTCCGAGCAGCTGGCTGG  
GCAACATAATTACTGCGGAATCTGATGGGATGCAAGCCCTGGTGGCAGCTG  
CTGAAGAAGCGCAGCGTGAAGTGGGAGTACTGTGATGGCTCTGGCTCCAGCTGC  
GGCTGAGAGCAGTACAGCAGCTCAGTTTCCACATCAAGGAGGGCTCTGGCGGA  
CATCGCTCCCAAGCTGGCAGCTGGCTCTTTCACAGCAGGAGGAGTGGCGCG  
AGAGCGTTCCTGTGGGGGGGATCTCATCAGCTCTGGCTGGATCTCTCTGGCG  
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CGCTCCAGGCGGATTTGGGAGGCGCCCTGGTGTGTCTGAAGCATGGCCGATGA  
CTTGGTGGGCACTACAGCTGGGGCTGGGCTGTGGACAGAGGATGTCCCGGT  
GTGTACACAAGGTTACCAACTACCTAGCTGGAATCTGTGACAAATGCGAGCG  
(SEQ ID NO:2)
```

- Data Matrix: A structured table consisting of records with fixed set of attributes. It is represented as an n by m matrix with n rows (objects) and m columns (attributes). The identity of the dataset is preserved if an entire column or row is shuffled.
- Spatial Data: Information including geographical location or 2D/3D physical spatial geometry.
- Temporal Data: Built-in support for handling time-related data.
- Graphs or networks: Topological structures that are defined by nodes (e.g. objects such as proteins or individuals) and edges (e.g. interactions such as protein-protein interactions (PPIs) or social network).
- Text: Natural language sequences
- Multi-modality Data: Combinations of multiple distinct data types, such as video(image, audio, transcripts), electronic health records (data matrices, images, text), spatial transcriptomics(spatial data, sequence, data matrix).

Introduction of Python:

What's programming?

The process of designing and building an executable computer program to accomplish a specific computing result or perform a task.



Communicating with the computer by correct language to complete tasks

Code: The specific message or instruction syntax transmitted to the computer $\approx$ the WhatsApp chat with friend, except you are texting with computer.

Python: An interpreted high-level general-purpose programming language which act as the communication bridge between human logic and hardware execution.

You wouldn't speak English to someone who doesn't understand it. In the same way, computers don't get human languages like Cantonese or English—so we use Python—one of the languages they actually speak—to communicate with them.



Language Translation Analogy

Computers only understand 0 and 1 (binary/machine-level instructions), while humans speak natural languages like English, so we can't talk to each other directly. Python acts as a middle ground: it's written in simple, human words that we can easily read, and a built-in translator converts it into instructions that the computer can run.

Basic Python Programming Mechanics

Python can be extended using specialized modules and packages for scientific computing, such as Numpy, Scipy, Pandas. (Need to type them first before using it).

```
import numpy
```

 (numpy can replace to other Plug-in)

Variables: Labeled boxes that hold data for you, so you don't need to keep retyping the same value again and again.

Assignment Syntax (=): To store a value, use = to put it inside the variable

e.g x=5 (store 5 in x)

```
a = [1,2,3,4,4,5,5,5,6,7,8,9]
```

Functions: Reusable mini programs that take in an input, do some math or logic, and hand back an output.

Parentheses (...): Used to run or call a function and feed it the details it needs to work.

Lists [...]: A way to group multiple items together into an ordered list.

```
a_mean = np.mean(a)
```

(Calculate the mean and store the returned value in a_mean)

If you want, you can change a_mean to any word, like Mum, Dad, handsome etc, using a_mean mainly for the clarity and readability.

Other example:

```
a_std = numpy.std(a)  
a_med = numpy.median(a)  
a_max = numpy.max(a)
```

Console Output (print()): Displays whatever you give it on the screen:

print("ABC") Outputs the exact word: ABC

print(ABC) Outputs the value inside: 100 (e.g ABC = 100)

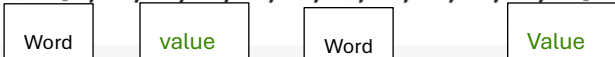
```
a_mean = numpy.mean(a)
```

(Value in a_mean and numpy.mean(a) is the same)



```
[19] print("The a array is ", a, "Its mean is ", a_mean)
```

The a array is [1, 2, 3, 4, 4, 5, 5, 5, 6, 7, 8, 9] Its mean is 4.916666666666667



```
[20] print("The a array is ", a, "Its mean is ", numpy.mean(a))
```

The a array is [1, 2, 3, 4, 4, 5, 5, 5, 6, 7, 8, 9] Its mean is 4.916666666666667